Graded Fault Response for Autonomous Driving via Continuous Sensor Health Monitoring and Risk/Capability-Aware MRM Supervision

Anonymous Submission

Abstract— Sensor-fault handling in automated driving stacks largely reduces to an immediate in-lane stop, while undetected degradation lets the vehicle keep driving with impaired perception. We propose a graded fault-response layer added onto an unmodified Autoware stack: (i) a behavior-tree monitor estimating per-sensor health as continuous scores (freshness, quality, physics) composed into capability scores, and (ii) an MRM supervisor selecting between shoulder/parking evacuation and deferral to the stock stop by fault type, severity, and remaining capability, with a conservative 3.0 m/s cap on evacuation movement. Across 1,862 valid closed-loop CARLA cases, detection reached 96–100% in every condition except one tolerated by design, with zero false activations, and degraded-state exposure fell from 238–275 m to 11–58 m. Verdicts rest on a dual scheme—an ego collision ledger and a pedestrian physics oracle—built after finding silent collision sensors. A pre-registered dose-response experiment (3 caps × 4 distances × 10 seeds) shows the cap causally moves the avoid/collide boundary inward (8 m: discordant pairs 10:0, exact McNemar $p = 0.002$); encounters in the post-detection deceleration transient remain at risk.

I. INTRODUCTION

Automated driving systems at SAE Level 4 and above cannot presuppose driver intervention: upon a fault they must execute a minimal risk maneuver (MRM) that reaches a minimal risk condition (MRC) on their own. ISO 26262 requires detection and response within the fault-tolerant time interval (FTTI), and SOTIF (ISO 21448) requires managing hazards from sensor performance limitations [1], [2], [3]. In the open-source stack Autoware, the default MRMs are in-lane stops (comfortable/emergency), and its stock diagnostics primarily target communication- and hardware-level anomalies such as topic silence and timeouts [9]. In the version and execution configuration evaluated in this study (Autoware 0.39.0), degradation faults whose values stay within valid ranges left driving to continue without intervention, and no supervisory layer existed that evaluates fault type and degree together with remaining maneuvering capability, selects between evacuation and stopping, and limits evacuation speed to match the fault state. That supervisory layer—not the shoulder maneuver itself—is the contribution of this work; the observation is version-specific, not a claim about Autoware in general.

An in-lane stop is simple and immediate, but the stopped vehicle blocks following traffic and creates secondary-collision exposure until the lane is cleared. Conversely, driving on with an undetected fault extends the distance exposed to hazards under degraded perception—in our unmonitored observations (one run per fault type) the stock stack completed 238–275 m of route with faults injected. The design space of fault response therefore has three axes: whether to stop now, how far to move, and—if moving—how fast.

Automating human-driver-like shoulder evacuation requires (i) distinguishing faults mild enough to permit moving from faults requiring an immediate stop; (ii) a continuous estimate of fault type and degree that makes this distinction possible (a binary normal/faulty state is insufficient); and (iii) managing the risk of moving itself in a degraded state—the stopping distance against a dart-out. We implement these as a continuous health score, risk- and capability-based state transitions, and a conservative capability-based speed cap.

Finally, simulation-based safety claims stand only on the reliability of the collision verdict itself. During our experiments we empirically uncovered a simulator collision-sensor configuration that remained silent; prompted by this, we built a dual adjudication scheme—a real-time ledger of the ego collision sensor and a pedestrian physics oracle—and based every safety conclusion on it.

Contributions. (1) A behavior-tree-based monitor that scores per-sensor freshness, quality, and physical consistency independently and composes them over eight-direction spatial sectors into perception/localization capability scores, published as continuous health. (2) A risk/capability-aware MRM supervisor that selects between shoulder/parking evacuation (MRM_MOVE) and deferral to the stock stop (MRM_DEFER) by health-score bands, chooses the refuge by a cost function over degraded-state exposure time, traffic-light observability, and sector deficits, and imposes a fixed 3.0 m/s cap (below the model-derived bound ≈3.89 m/s from $d = v\tau + v^2/2a$) on evacuation movement. (3) A dual collision-adjudication methodology (ego RAW-HIT ledger + pedestrian physics oracle) with an explicit catalogue of verdict pitfalls in runner exit labels and official infraction records. (4) Encounter-level and large-matrix validation totalling 1,862 valid closed-loop cases, including a pre-registered dose-response experiment that verifies the causal effect of the cap on settled encounters, and transient-vs-settled observations locating residual evacuation risk in the post-detection transient.

The novelty is not in the individual techniques (CUSUM, residual monitoring, noisy-OR fusion,

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stopping-distance model—each known), but in the integrated supervisory layer spanning detection through selection and speed, and in the closed-loop demonstration whose verdict reliability is secured at the methodological level.

II. RELATED WORK

Standards. Functional-safety and SOTIF frameworks [1], [2], [3], [4] consistently require runtime monitoring and responses to degradation. ISO 23793-1 [5] specifies an MRM type taxonomy and the principle of situation-dependent type selection but covers only straight-line and in-lane stops (shoulder stops are being standardized as Part 2). The industrial driverless-truck standard ISO 3691-4 [6] prescribes "limited speed for limited capability"; our conservative low-speed MRM movement transfers this principle to on-road MRM. None of these standards provides a method to continuously estimate fault kind and degree and to select response and speed accordingly.

Fault detection and degradation models. Classical FDI develops residual generation/decision [7] and sequential change detection [8]; our LiDAR quality channel borrows CUSUM. These theories end at detection—maneuver selection after detection is out of scope. Our LiDAR sparsification model follows field measurements of soiling-induced range reduction [10], and the clutter model follows rain-scattering attenuation and near-range-biased spurious returns [11], [12]; camera failure modes follow [13], IMU noise/bias separation [14], GNSS integrity and multipath-type correlated bias [15], [16], [17].

Fallback and MRM planning. Fail-safe trajectory planning [18], [19] maintains stop trajectories and online verification but takes the fault assessment and response choice as given. Emzivat et al. [20] proposed a fallback for impaired monitoring capability; Reschka and Maurer [21] pointed out that an in-lane stop is not always a safe state—the concern we start from. Closest to us, Xue et al. [22], [23] presuppose a forward-sensor fault and plan safe-zone stops under MPC constraints, and Tang et al. [24] adaptively select MRM behaviors under multiple sensor faults using HAZOP risk analysis with TTC/THW evaluation. They differ from this work in that (i) the sensor-fault state is presupposed rather than estimated online; (ii) selection uses surrounding-traffic TTC/THW, whereas we combine risk grace time with remaining perception/localization capability and additionally limit the evacuation speed itself; (iii) validation checks representative scenarios, whereas we verify 1,862 valid cases on a real stack with encounter experiments whose verdicts are methodologically secured. Capability-based self-monitoring (skill/ability graphs) [25] models capabilities for online decisions; our sector-capability composition couples capability scores directly to MRM level selection and the speed cap. Behavior trees [26] serve as our implementation structure for parallel composition and execution tracing.

Simulation-based evaluation. CARLA [27] hosts stack-level fault-injection studies (e.g., DriveFI [28], AV-FUZZER [29]) that inject faults to find vulnerabilities—the opposite direction from ours, where the stack itself detects, grades, and responds to injected faults. Our infrastructure builds on AWCL [30], the TUM Autoware–CARLA leaderboard coupling. We found no prior report addressing the reliability of simulated collision verdicts (sensor silence, label misclassification) at the methodology level.

III. PROPOSED SYSTEM

A. Architecture

The system consists of two independent add-on nodes (Fig. 1). The monitoring module inspects each sensor at 10 Hz on a behavior-tree basis, computing a per-sensor health score in $[0, 1]$ from three detector families: timeout (freshness), quality (noise, valid ratio), and physics (residual, bias). The MRM supervisor subscribes to the scores and runs a state machine: for evacuation it changes the route via the Autoware AD API and imposes a speed cap on the planning layer; for deferral it triggers the stock mrm_handler stop through the diagnostic graph. Stopping-maneuver execution is delegated to the stock stack (goal_planner, mrm_handler); the main stack is not modified. Reception accumulates at native sensor rates; evaluation runs at 10 Hz, consistent with AUTOSAR watchdog-style periodic supervision.

B. Sensor Scoring

Freshness. For elapsed time age(t) since last reception, $s_{\text{fresh}} = \max(0,\, 1 - \text{age}/\tau_{\text{detect}})$ decays linearly—so the score can cross the downstream band boundary before the budget is exhausted, enabling early transition under complete silence. A k-miss rule penalizes early when nothing arrives for $3\times$ the moving median of the reception interval.

Quality. For LiDAR, a Gaussian noise level is recovered from neighboring-point range differences, $\hat{\sigma} = \overline{|\Delta r|} \cdot \sqrt{\pi}/2$ (samples with $|\Delta r| < 1$ m), referenced to a boot-time baseline that refuses to track abrupt rises (so the fault cannot become its own baseline); the score is $\exp(-\max(0, \hat{\sigma} - \text{base})/0.1)$ times a valid-point-ratio score, with a parallel CUSUM sequential test so detection delay is inversely proportional to fault magnitude. Camera quality applies the same structure on inter-frame pixel differences (no cumulative test, since pixel difference rises naturally with scene motion). Sparsification is caught by the valid-ratio channel and clutter by the range-discontinuity channel.

Localization (three channels). (a) a residual channel on the GNSS–EKF distance (sensitive to noise and multipath-type correlated bias); (b) a bias channel comparing 3-s means of IMU and EKF yaw rates (normal turning cancels; gyro bias remains on the IMU side); (c) a silence channel that decays the score when GNSS

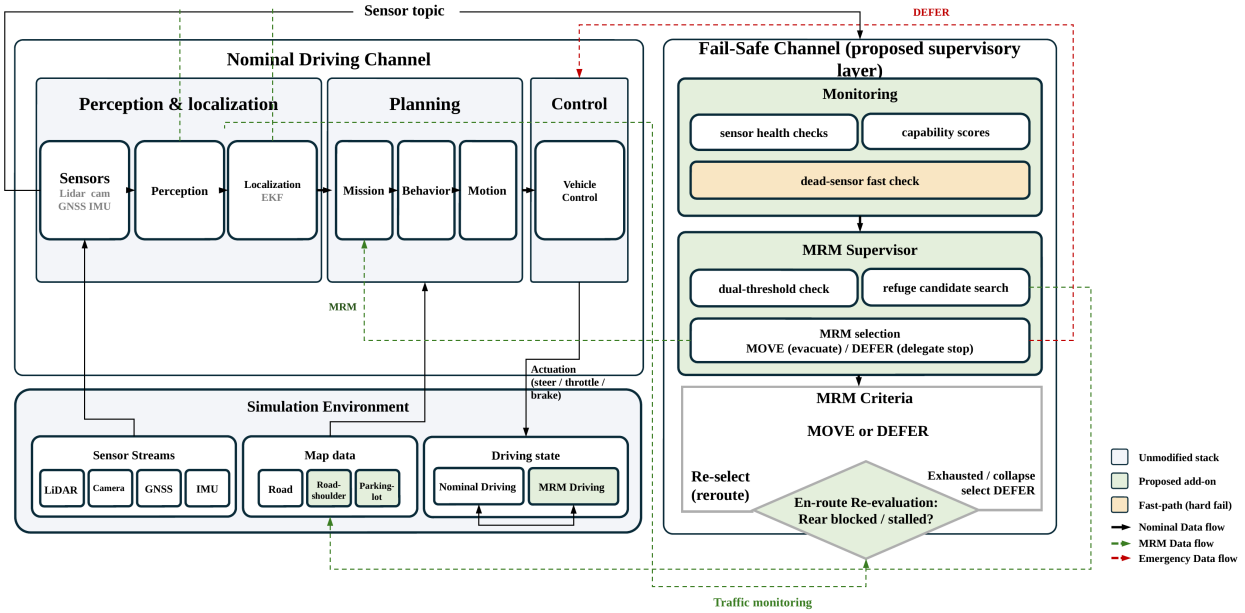


Fig. 1. Dual-channel structure. Left: the nominal driving channel of unmodified Autoware (perception & localization → planning → control); right: the proposed supervisory layer. Monitoring subscribes read-only to sensor topics and produces health/capability scores; the decision stage selects MOVE (evacuate: reroute with a speed cap) or DEFER (inject a diagnostic ERROR to trigger the stock stop), re-evaluating rear occupancy and stalling while moving and falling back to DEFER when candidates are exhausted. Bottom: the CARLA closed loop; refuge-candidate search draws on manually annotated map data.

input stops ≥1 s while driving—covering the residual channel's blind spot of freezing at the last value.

C. Sector and Capability Synthesis

Sensor scores map to eight directional sectors by mounting direction and field of view and fuse as $s_{\text{sector}} = 1 - \prod_i (1 - s_i)$; perception capability is a sector-weighted sum (front-weighted), combined with traffic-light observability at 0.6:0.4; the system health is min(perception, cap_{loc})—the weakest-link principle.

D. Risk-Coupled Detection Budget

The freshness decay constant is coupled to driving risk: $\tau_{\text{detect}}(t) = \min(\tau_{\text{fresh}}, \text{TTC}_{\text{cons}}(t))$, where TTC_{cons} is the minimum of nearest-obstacle TTC and path-crossing time with a fail-conservative hold when the TTC source itself is unhealthy. The design intent is $T_{\text{detect}} + T_{\text{response}} + T_{\text{margin}} \leq T_{\text{hazard}}$; the implementation is a risk-adaptive heuristic approximation of this budget, not a formal guarantee (explicit FRTI/margin reservation is future work). For degradation faults, the requirement is a severity curve $T_{\text{detect}}(\sigma) \leq T_{\text{hazard}}(\sigma)$: small faults are caught later but become hazardous later.

E. MRM Supervisor

Dual thresholds on health (upper 0.7, lower 0.3) span three bands: NORMAL, evacuation (MRM_MOVE), and deferral (MRM_DEFER, on floor entry or localization-confidence collapse). Decay constants were designed so a wide insensitive gap lies between faulted and normal scores; behavior follows the band structure, not precise threshold placement. Timing splits by evidence type: degradation faults are statistical evidence and require sustained confirmation (1.0 s band entry; 2.0 s at the floor since deferral is terminal; a 5.0 s recovery grace that a 0.5–5.0 s sensitivity sweep left unchanged), whereas hard fails (silence past the 850 ms freshness deadline) already carry accumulated evidence, so confirmations are skipped and unrecoverable conditions defer immediately. Selection is not frozen while moving: sustained rear occupancy demotes reversing parking candidates to a shoulder stop, stalls get bounded retries, and candidate exhaustion or en-route health collapse falls back to DEFER.

Refuge selection minimizes

$$J(c) = w_t\,\eta(c) + w_{tl}\,n_{tl}(c)\,g(tl) + w_{sec}\sum(1-s_{\text{req}}) + w_{loc}\,\eta(c)(1-\text{loc}) \quad (1)$$

where $\eta(c)$ is degraded-state exposure time (route time + expected signal waits)—risk accumulates per unit time, not distance. Route times and traffic-light counts are computed on an in-house road-network graph parsed directly from the lanelet2 map, keeping the decision layer outside the influence of main-stack faults. The key safeguard is the traffic-light observability gate: a candidate whose route passes signals no healthy camera can read is excluded with $J = \infty$. Shoulder/parking candidates are manually annotated map attributes (16 shoulders, 4 lots on the test map).

F. Conservative Low-Speed MRM Movement (Speed Cap)

Evacuation is movement under degraded perception, so stopping capability against a dart-out dominates. With reaction time τ and deceleration a, stopping from speed v requires

$$d_{\text{safe}}(v) = v\tau + \frac{v^2}{2a} + d_{\text{margin}} \leq D_{\text{avail}}. \tag{2}$$

Here $\tau \approx 1.5$ s, $a \approx 3.5$ m/s^2 are nominal lumped values calibrated to the measured ladder boundary—a design approximation for choosing the cap policy value, not a predictive safety model. On entering MRM_MOVE the supervisor imposes a fixed 3.0 m/s cap, chosen inside the band whose upper end is the model-inverted cap that still stops within an 8 m dart-out (≈3.9 m/s) and whose lower end bounds evacuation-completion time; no optimization within the band is claimed. The cap is delivered to the input of Autoware's velocity-limit arbiter and its delivery is verified by measured vehicle speed, not publish records—an interface mismatch that silently dropped the cap in earlier campaign runs was discovered exactly this way and corrected (Sec. V-D).

IV. EXPERIMENTAL SETUP

A. Environment and Campaign Matrix

The closed loop couples CARLA 0.9.16 with Autoware 0.39.0 over the AWCL bridge [30] on CARLA Town07 (custom vector map with annotated refuges; speed limit 15 km/h). The simulator is fully restarted for every case. A case is defined by {method, fault, severity, traffic, repetition, injection time}; methods are ours (proposed supervisor), baseline (identical detection, deferral to the stock stop—isolating response strategy from detection), and mrm-off (passive scoring only). The main campaign is fault × severity × traffic density (none/defensive/aggressive) × method with 1,131 valid cases (39 short of the 1,170 plan, all in high-severity LiDAR clutter); a navigation-extension matrix with identical structure adds four navigation faults (611 valid), and encounter experiments complete the 1,862 total. The fault model injects inside the agent's sensor pipeline: camera death/noise (cf/cn), LiDAR death/sparsification/clutter (lf/lso/lcl, physical models [10], [11], [12]), GNSS noise/correlated bias/outage (gn/gb/gf [16]), and IMU noise/bias/drift (imn/imb/imd), with severities low/mid/high where applicable; injection times and fault waveforms are seed-fixed for pairing.

B. Collision Adjudication

The initial implementation adjudicated collisions with a collision sensor attached to the pedestrian; measurement showed this sensor never fired in CARLA, and a listener death additionally lost separation records, so non-intervention-arm cases were being misjudged as avoided. All affected data were discarded and re-run. The reconstructed system has two layers: (i) primary: a real-time ledger (RAW-HIT) written by the patched callback of the ego's sensor.other.collision the moment a collision occurs, with counterpart actor identifiers (endpoint-structure grazes are classified separately)—unlike official infraction tallies it survives early/forced termination; (ii) audit: a pedestrian physics oracle (kinematic anomalies: flung >4.5 m/s or a >5 m jump within 1 s) and the per-tick separation stream. Runner exit labels are refused as evidence: the collision label can include endpoint contacts, the hazardous-stop label can include post-collision stops, and official infractions are saved only on natural termination. All encounter-level figures in this paper are ledger-based. Latency figures use simulation time (actor destruction stalls the simulator ∼1.6 s; wall clock overestimates).

C. Encounter Experiments

A decoupled injector process spawns pedestrian dart-out crossings at the right lane edge with per-seed fixed walking speed, lateral jitter, and fault waveform. Designs: (i) a fault-free distance ladder at 4.2 m/s, spawn distances $\{14, 11, 8, 6, 5\}$ m, n=4 each; (ii) the same ladder blinded (lso-high); (iii) an 8 m encounter comparison under gn-mid (EKF input contaminated, perception intact—so all arms share detection capability); (iv) a low-speed ladder at 3.0 m/s cruise; and (v) a dose-response experiment: caps $\{4.2, 3.6, 3.0\}$ m/s × distances $\{6, 7, 8, 9\}$ m × 10 seeds = 120 runs, launched by a settlement trigger—the pedestrian spawns D m ahead only when the supervisor is in MRM_MOVE and measured speed has stayed in the cap-specific band for 1.0 s—so an encounter under an actually holding cap is guaranteed at trigger level. The primary endpoint (8 m, paired 4.2 vs 3.0 exact McNemar) and a manipulation-check adoption rule (first-tick speed in band ±0.15 m/s, ahead-of-ego geometry, launch within 420 s; violations re-run blind to outcome) were pre-specified before data collection; sample size was fixed at 120 adopted runs.

V. RESULTS

A. Detection Performance

Table I summarizes the campaign. Detection rates are 96–100% in 19 of 20 fault conditions—median latencies ∼1.0 s for hard fails (ledger, simulation time) and 2.1–8.6 s for degradation faults (wall clock; different clock domains, not directly comparable)—with zero false activations in 45 fault-free control cases. The sole exception, mid-severity LiDAR sparsification (2%), lies by design in the tolerated band: at that dropout level ∼55 m effective range remains and the supervisor deliberately continues driving. Since tolerance-as-designed cannot yet be separated from quality-channel insensitivity, we read the table as intervention eligibility rather than fault-existence detection. Notable structure: the IMU drift latency is inversely related to ramp slope (8.6 vs 5.4 s)—small faults are caught later but become dangerous

TABLE I

Detection rate and median latency per fault condition (proposed detectors; n=45 per condition unless noted). †: navigation-extension matrix. *: ledger-based simulation time; others wall clock.

Fault	Sev.	Det. rate	Median lat. (s)
cf camera death	–	1.00	0.98*
cn camera noise	mid/high	0.98/1.00	3.3/3.2
lf LiDAR death	–	1.00	1.00*
lso LiDAR sparsif.	mid	0.02	– (tolerated band)
lso LiDAR sparsif.	high	0.98	5.0
lcl LiDAR clutter	mid/high[a]	1.00/0.96	4.8/4.5
gn GNSS noise	mid/high	1.00/1.00	2.1/4.4
imn IMU noise	mid/high	1.00/0.98	4.8/3.7
gb† GNSS corr. bias	mid/high	1.00/1.00	3.0/2.3
gf† GNSS outage	–	1.00	3.2 (driving; bimodal)
imb† IMU gyro bias	mid/high	1.00/1.00	3.6/3.2
imd† IMU bias ramp	mid/high	1.00/1.00	8.6/5.4
nf fault-free control	–	0 false activations / 45	

[a] lcl-high n=26 (final batch halted; accounted).

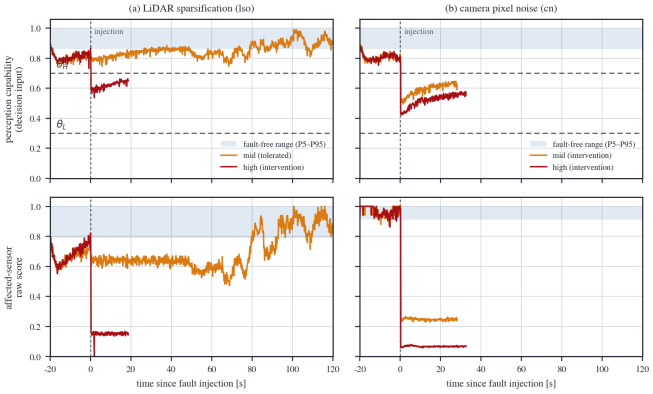


Fig. 2. Injection-aligned health traces (top: the perception capability score feeding the 0.7/0.3 transitions; bottom: affected sensor's raw score) for LiDAR sparsification and camera noise. Fault-free runs stay in the normal band (measured min 0.71); mid severity stays in the tolerated band (intended non-intervention of lso-mid); high severity departs immediately and triggers intervention—the band separation underlying the graded three-regime response.

later; GNSS-outage latency is bimodal (median ∼3 s while driving; 21–40 s when the outage coincides with a red-light stop, an intended delay since a stationary vehicle is not a risk exposure); and deferral stops split by family (GNSS → comfortable stops, IMU → hard stops—corrupted rotational sense demotes trajectory following).

Behind the binary rates lies the continuous signal (Fig. 2): the decision variable $\min(\text{perception}, \text{cap}_{\text{loc}})$ separates into distinct bands by fault intensity, which is the basis of the graded normal/evacuate/defer structure.

B. Graded Evacuation Behavior and Exposure

For faults preserving the capability to move, the proposal evacuated while the baseline (by definition) stopped in lane: camera death 16/45 shoulder evacuations (27 intended deferrals, 2 maneuver incompletions), camera noise 25/90, GNSS noise 42/90. Severity determined behavior most sharply in the navigation family:

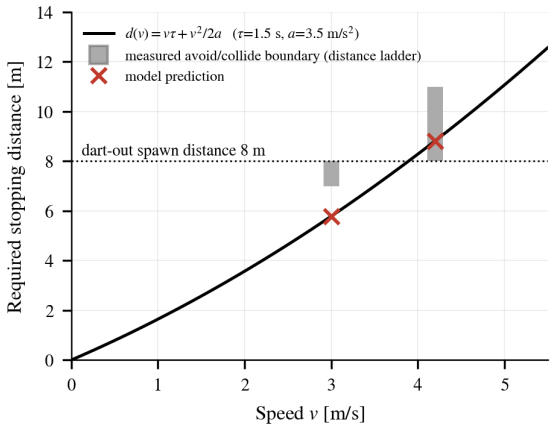


Fig. 3. Stopping-distance model $d = v\tau + v^2/2a$ (τ=1.5 s, a=3.5 m/s^2) vs. measured avoid/collide boundaries (shaded bands). At 4.2 m/s the measured 8–11 m matches the model's 8.8 m; at 3.0 m/s the model's 5.79 m is a lower-bound approximation of the measured 7–8 m. Dashed line: the 8 m spawn used in subsequent experiments.

gn-mid produced 40/45 shoulder evacuations, whereas gn-high converged to deferral stops in 42/45—movement is abandoned when localization confidence falls below the movement minimum. Conversely, lso-mid continued driving deliberately (response proportional to degree, not stop-at-every-anomaly). Unmonitored (mrm-off, one run per type) the stack completed 238–275 m in the faulted state; with monitoring, exposure became a policy variable: 32–58 m for detect+evacuate (camera/navigation families), 9–18 m for detect+stop, and 11–12 m for LiDAR clutter. The in-lane stop minimizes exposure at the cost of lane occupancy; evacuation accepts tens of meters of controlled exposure for lane clearance and an off-lane refuge.

C. Avoid/Collide Boundary and the 8 m Encounter

The fault-free ladder at 4.2 m/s produced a clean staircase: 14 and 11 m avoided 4/4 each; 8, 6, 5 m collided 4/4 each—boundary in 8–11 m, agreeing with Eq. (2)'s prediction of 8.8 m (Fig. 3; the same data calibrated the model, so this is self-consistency, not independent validation). Blinded (lso-high), the stock stack froze mid-road before the encounter at 14/11 m and collided at ≤8 m: an unmonitored degraded vehicle either stands still or strikes—neither an MRC.

Under gn-mid (perception intact, so all arms share detection capability), an 8 m dart-out gave: stock at 4.2 m/s collided 10/10; the proposed low-speed settled cruise avoided 8/8 and completed shoulder evacuation (auxiliary Fisher $p = 2.3 \times 10^{-5}$); and—critically—the same proposed policy collided 2/2 when the encounter fell in the deceleration transient (encounter speeds 3.5–3.8 m/s). Protection is effective only after deceleration completes; residual risk concentrates in the post-detection transient. The low-speed ladder placed the

TABLE II

Dose-response experiment: collisions per cell (n=10 seeds; ledger adjudication; settled encounters only).

Cap	6 m	7 m	8 m	9 m	Boundary
4.2 m/s (control)	10/10	10/10	10/10	10/10	beyond 9 m
3.6 m/s	10/10	10/10	10/10	2/10	∼9 m
3.0 m/s (policy)	8/10	5/10	0/10	0/10	∼7 m

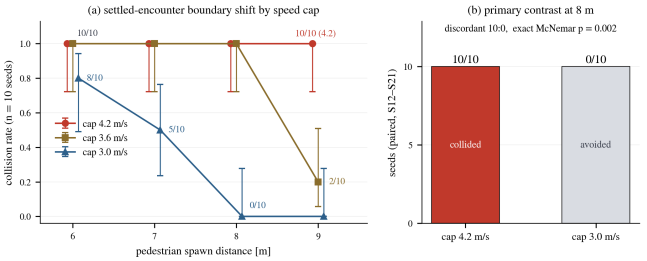


Fig. 4. Dose-response experiment (120 runs, ledger adjudication). (a) Collision rate vs. spawn distance per cap (Wilson 95% CI)—the boundary moves inward monotonically as the cap is lowered. (b) Pre-specified primary contrast at 8 m: 10/10 vs. 0/10, discordant pairs 10:0, exact McNemar $p = 0.002$.

3.0 m/s boundary at 7–8 m (8 m: 8/8 avoided; 7, 6, 5 m: 2/2 collisions each).

D. Dose-Response Verification of the Speed Cap

Because the two ladders used different launch protocols, the cap effect was verified in a controlled preregistered experiment on settled encounters only, after fixing a cap-delivery interface defect (the earlier "capped" runs were receiving no cap—publish records are not delivery; measured speed is). All 120 adopted runs passed the manipulation check (first-tick encounter speeds: 4.05–4.13, 3.37–3.58, 2.60–3.07 m/s per arm; five failing runs were rejected blind to outcome and re-run). Results (Table II, Fig. 4) are monotone along both axes. At the pre-specified primary endpoint (8 m), cap 4.2 collided in all ten seeds and cap 3.0 in none—discordant pairs 10:0, exact McNemar two-sided $p = 0.002$: once the cap actually holds, its collision-reduction effect is detected with unanimous seed-level directionality. The boundary moves inward monotonically: beyond 9 m (4.2), ∼9 m (3.6, 2/10), ∼7 m (3.0, 5/10 at 7 m, 0/10 at ≥8 m), mutually corroborating the low-speed ladder's 7–8 m interval. Interpretation limits: the causal conclusion is restricted to the settled regime (the cap does not remove the transient; it creates the post-settling margin), no optimality is claimed for 3.0 m/s, and cells are n=10, single fault (gn-mid), single corridor.

E. Hard-Fail Decision Latency

Hard fails are decided by an evidence-aware fast path (skipping noise-oriented debounce and futile recovery holds) against an internal 1.5 s detection-to-decision design deadline (not an FTTI-compliance claim). Over the full camera/LiDAR-death matrices, ledger-based simulation-time latencies gave medians ≈1.0 s, P95 1.05–1.15 s, maximum 1.20 s, zero violations, with MRM attainment maintained in full.

VI. DISCUSSION AND LIMITATIONS

Stop vs. evacuation is a temporal trade-off. In this sample the in-lane stop showed the lowest collision rate even under transient-window launches, at the price of sustained lane occupancy; evacuation accepts controlled moving exposure and residual transient risk, but on success clears the lane and reaches an off-lane refuge (lane clearance indicates relinquished occupancy, not proven final-position safety—rear-approach risk at the shoulder was not compared). Neither is unconditionally superior; the choice should follow the fault's temporal structure—concrete selection variables for the ISO 23793-1 framework.

Residual risk is temporally localized. All evacuation-arm collisions occurred before deceleration completed (transient 2/2) and none after settling (0/8; all settled dose-response encounters avoided at the policy cap's boundary or beyond). Distinguish the post-detection transient (t_{detect} to settled speed) from the overall fault-response risk window (fault onset to settled speed, $T_{\text{detect}} + T_{\text{decel}}$): the results support—observationally, not as a causal manipulation of detection timing—that earlier detection and faster deceleration shrink this window, i.e., widen the margin within the FTTI.

Adjudication lessons generalize. "Zero collisions" is meaningful only after the verdict channel is proven alive: positive-control verification of collision sensors, physics-oracle cross-checks, and refusal to read absence of a record as absence of an event. The silently undelivered speed cap carries the same lesson—actuation must be verified by downstream effect. Both failure modes arise structurally in automated large-scale experiments; the real-time ledger was the effective remedy.

Limitations. Single map, single 15 km/h speed limit, two launch corridors: the numeric boundaries (8–11 m, 7–8 m) and dose-response numbers are low-speed-ODD values; generalizable claims are structural (the stopping-distance dependence of the boundary, transient risk concentration, the stop/evacuation trade-off). Autoware observations are version-specific (0.39.0). Dose-response cells are n=10 with wide CIs; testing was restricted to the single pre-specified endpoint. Campaign-level collision-free status predates the ledger (no pedestrian injections there, but stated as a methodological-consistency gap). The dynamic detection budget was not ablated against a fixed deadline; hazard-time-referenced deadline compliance ($T_{\text{hazard}} - T_{\text{detect}}$) requires an instrumentation extension; the health-proportional variable cap is implemented but unevaluated; no real-vehicle validation.

VII. CONCLUSION

We proposed and validated, in an Autoware–CARLA closed loop of 1,862 valid cases, a supervisory add-on layer that estimates sensor faults as continuous

health scores and selects MRMs in a graded manner—evacuation vs. stop deferral by fault type, severity, and remaining capability, with a conservative speed cap on evacuation movement. Three conclusions carry beyond the numbers. First, shoulder evacuation is no panacea: stopping and evacuation form a temporal trade-off between immediate collision risk and lane clearance, and the choice should follow the fault's temporal structure. Second, the residual risk of evacuation concentrates in the post-detection transient, while settled low speed verifiably moves the avoidance boundary inward (transient 2/2 vs. settled 0/8; dose-response 10:0, $p = 0.002$)—supporting earlier detection and faster deceleration as means of shrinking the fault-response risk window within the FTTI. Third, simulation-based safety claims hold only on a validated adjudication framework; the dual ledger/physics-oracle scheme and the catalogued verdict pitfalls are reusable methodology for any simulated MRM comparison. Future work includes high-speed ODDs and multiple maps, formalizing the detection budget, evaluating the health-proportional cap, and real-vehicle validation.

References

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